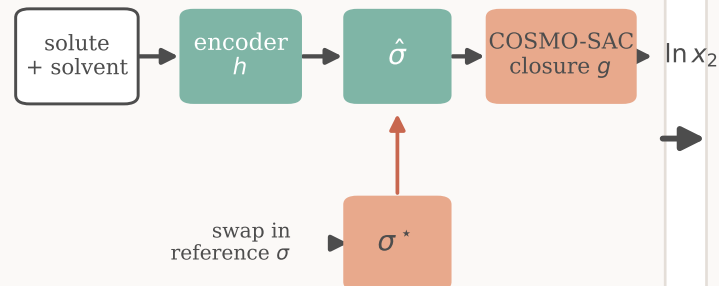


When do reference physical inputs help a learned solubility model?

1. The paradox

fixed, no fitted parameters



Premise: truer physical inputs $\rightarrow$ better.

Here, the opposite. So which is wrong — the closure, or the inputs?

2. The measurement

the reference input has an external oracle (VT-2005),

so the reference-input error splits exactly:

$$B = B_{\text{closure}} + B_{\text{insuff}}$$



$$B_{\text{closure}} > B_{\text{insuff}} \quad (P_{\text{boot}} \approx 0.78)$$

The ceiling is the CLOSURE, not the inputs.

3. The lever

Raise closure fidelity and grounding flips from hurt to help.



closure fidelity $\rightarrow$

ours
(2002)

2010/
dsp

TeNNet

*(latent supervision is a second lever,
left to future work)*

What the paper establishes, graded by evidence

CERTIFIED

- Paradox: reference σ makes it worse (3 seeds, $n = 5608$)
- Exact split $B = B_{\text{closure}} + B_{\text{insuff}}$ (Lemma 2)
- Closure = NIST COSMO-SAC-2002 to $0.003 \ln \gamma$

LIKELY

- Closure binds on low- γ set: $P_{\text{boot}} \approx 0.78$
- A modern closure (2010/dsp) removes the penalty
- Latent = low-rank, transferable surrogate ($n = 44$)

EXPLORATORY

- Transfer to the finite-composition regime
- γ -stratified strengthening (estimator-dependent)
- pKa second closure as a fidelity cross-check